Let $x \neq 2m\pi$ $(m \in \mathbb{Z})$ be a real number. Observe that

$$\sum_{k=1}^{n} e^{ikx} = e^{ix}\cdot\sum_{k=0}^{n-1} e^{ikx} = e^{ix}\cdot\frac{1-e^{inx}}{1-e^{ix}} = e^{ix}\cdot\frac{e^{\frac{1}{2}inx}}{e^{\frac{1}{2}ix}}\cdot\frac{e^{\frac{1}{2}inx}-e^{-\frac{1}{2}inx}}{e^{\frac{1}{2}ix}-e^{-\frac{1}{2}ix}}$$

$$= e^{\frac{ix}{2}\cdot(n+1)}\cdot\frac{\sin\frac{nx}{2}}{\sin\frac{x}{2}}$$

Since $e^{ikx} = \cos kx + i\cdot\sin kx$ and

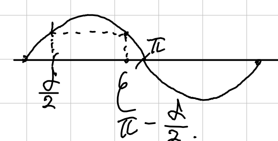
$$e^{\frac{ix}{2}(n+1)} = \cos\frac{(n+1)}{2}x + i\cdot\sin\frac{(n+1)}{2}x$$

So therefore

$$\sum_{k=1}^{n}\cos kx = \frac{\sin\frac{nx}{2}}{\sin\frac{x}{2}}\cdot\cos\frac{n+1}{2}\cdot x \quad \text{and} \quad \sum_{k=1}^{n}\sin kx = \frac{\sin\frac{nx}{2}}{\sin\frac{x}{2}}\cdot\sin\frac{n+1}{2}\cdot x$$

Now, for fixed $0<\delta<\pi$, given $x \in [\delta, 2\pi-\delta]$,

$$\left|\sum_{k=1}^{n}\cos kx\right|, \left|\sum_{k=1}^{n}\sin kx\right| \leq \left|\frac{1}{\sin\frac{x}{2}}\right| \leq \left|\frac{1}{\sin\frac{\delta}{2}}\right|$$



Consequently, we obtain a Theorem:

Thm. Suppose that $\{a_n\}$ converges to 0, and $\sum |a_n - a_{n-1}|$ converges.
Given $\delta \in (0,\pi)$, $\sum_{n=1}^{\infty} a_n\cdot\sin nx$ and $\sum_{n=1}^{\infty} a_n\cdot\cos nx$ converges uniformly on $[\delta, 2\pi-\delta]$.